

WASP-36b

R-2343

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_130227 · created 2026-08-19T00:33:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456350.82052 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.018
Transit depth (Rp/Rs)^2	2.79 +/- 0.6 [%]
Orbital Inclination (inc)	83.85 +/- 1.64
Airmass coefficient 1 (a1)	1.295 +/- 0.037
Airmass coefficient 2 (a2)	-0.131 +/- 0.012
Scatter in the residuals of the lightcurve fit is	2.81 %
Best Comparison Star	None
Optimal Aperture	3.48
Optimal Annulus	11.23
Transit Duration (day)	0.082 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.018 vs 0.1368 (deviation 1.41σ)
Duration fit vs published	0.082 d vs 0.0758 d (deviation 0.52σ)
Mid-time O–C	1.3 min
Depth SNR	7.8
Residual scatter	2.81 %

Light curve

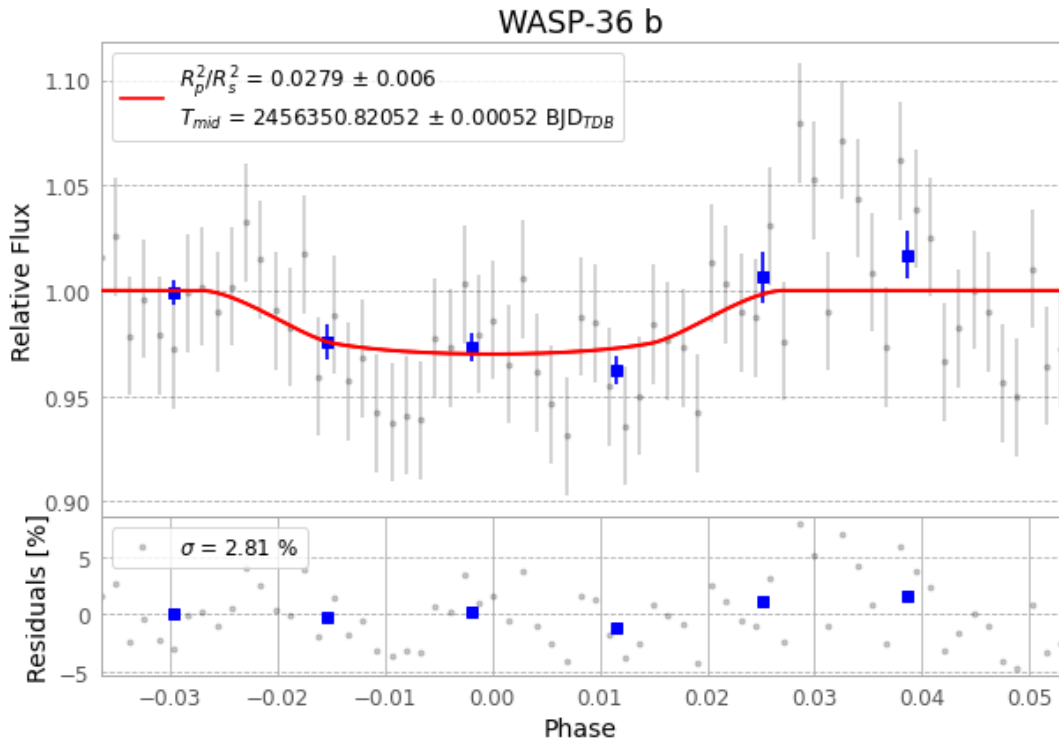


Figure 1 — FinalLightCurve_WASP-36 b_2013-02-26.png (SHA-256 003d69fd2a808410...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 332], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ec3b27b10ac11bc3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

68 FITS frames (45 MB), WASP-36130227061213.FITS → WASP-36130227093312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130227.log (e878d8a8c3fb...), FinalLightCurve_WASP-36 b_2013-02-26.png (003d69fd2a80...), FinalLightCurve_WASP-36 b_2013-02-26.csv (c12abd06f995...)

Compute resources

Wall time 112.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:33Z.